Short-Term Solar Power Forecasting

Comparing FTDNN, NARX, and LSTM

Evan Burk • Deep Learning Final Project

Why this project matters

Solar output changes quickly, so one-hour forecasting is operationally useful.

Grid and storage decisions

Better short-term forecasts help with charging, load planning, and renewable integration.

A real dataset

The project uses self-collected time-series logs rather than a benchmark dataset, which makes the result more practical.

A good model comparison

Comparing LSTM against simpler FTDNN and NARX baselines, including linear variants.

Project framing

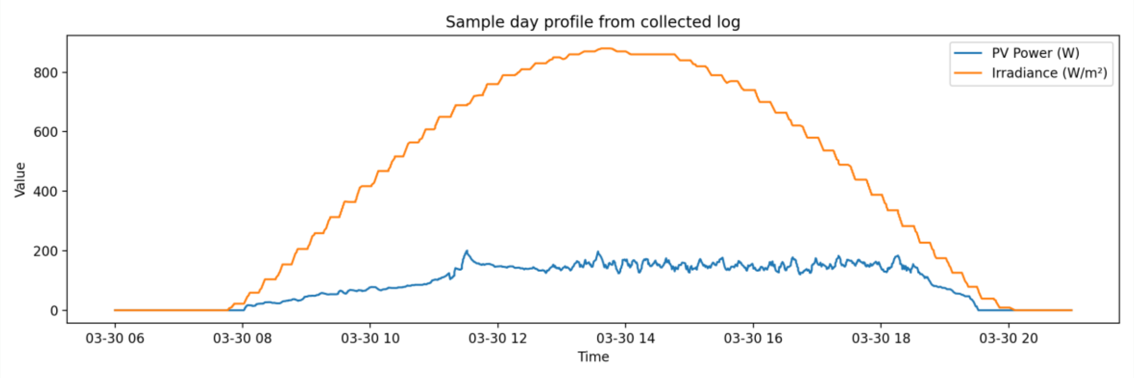
- Target: predict future PV power output 60 minutes ahead.
- Approach: compare feedforward, autoregressive, and recurrent sequence models in PyTorch.
- Success criteria: beat persistence and compare models with MAE, RMSE, and R^2.

Project setting and data source

Daily logs from the solar lab are merged into one chronological time series.

What the logs contain

Panel voltage/current/power, battery measurements, weather, irradiance, UV, and derived efficiency. The right plot shows a sample day profile from a collected log.



- CSV files are written daily as gpg_YYYY-MM-DD.csv in a logs folder.
- The modeling code reads every daily file and sorts the merged data by timestamp.
- This creates a realistic forecasting setup from the same data stream used in the lab.

Dataset snapshot

The final run used 28,173 rows after preprocessing and 19 input features.

ROWS AFTER PREPROCESSING	FEATURES	LOOKBACK	HORIZON
28,173	19	120 min	+60 min

Electrical / battery signals

PV_Voltage_V
PV_Current_A
Battery_Voltage_V
Battery_Current_A
Battery_Power_W
Avg_Power_to_Battery_W

Environmental signals

Battery_Temp_C
Air_Temp_C
Humidity_pct
Air_Pressure_inHg
Wind_Speed_ms
Wind_Direction_deg
Solar_Irradiance_Wm2
UV_Index

Derived + seasonal signals

Est_Efficiency
mod_sin / mod_cos
doy_sin / doy_cos

Forecasting task

Each sample uses a two-hour history window to predict PV power one hour into the future.

Model input window

120 minutes of past signals
PV + battery + weather + time features

Prediction target

PV_Power_W at t + 60 minutes

Why chronological splitting matters

Random shuffling would leak future information into the past. Time-series data must be split in order.

Why a horizon of +60 min matters

A one-hour horizon is long enough to be useful, but short enough that recent conditions still carry signal.

Baseline check

Every learned model is compared to persistence: future equals the latest observed power.

Preprocessing and chronological split

The model only sees past information when predicting the future.

- Parse timestamps and sort all daily logs in time order.
- Drop unusable rows and enforce a maximum 2-minute gap so outages do not appear continuous.
- Build sliding windows only from past data.
- Scale features using training-set statistics only.
- Use early stopping based on validation loss.

Why this matters

This setup mirrors the real forecasting problem: the test set happens strictly later in time than training, so reported performance is more trustworthy than a shuffled split.

Baseline and evaluation metrics

A model only matters if it improves on a simple forecast.

Persistence baseline

Predict PV power at t + 60 min using the latest observed PV power. It is simple, strong, and easy to beat only if the model learns real temporal structure.

MAE

Average absolute error. Easy to interpret in watts.

RMSE

Penalizes larger misses more strongly than MAE.

R^2

Explained variance relative to a constant-mean predictor.

Why compare FTDNN, NARX, and LSTM?

They represent three different ways to model sequence information.

FTDNN

Focused Time Delay Neural Network

- Feedforward network with delayed inputs
- Simple and parallelizable
- Good baseline for finite-memory sequence prediction
- Tested here in linear and nonlinear forms

NARX

Nonlinear Autoregressive with Exogenous Inputs

- Uses past inputs and past outputs
- Natural fit for dynamic systems
- Can be trained in series-parallel form
- Also tested in linear and nonlinear forms

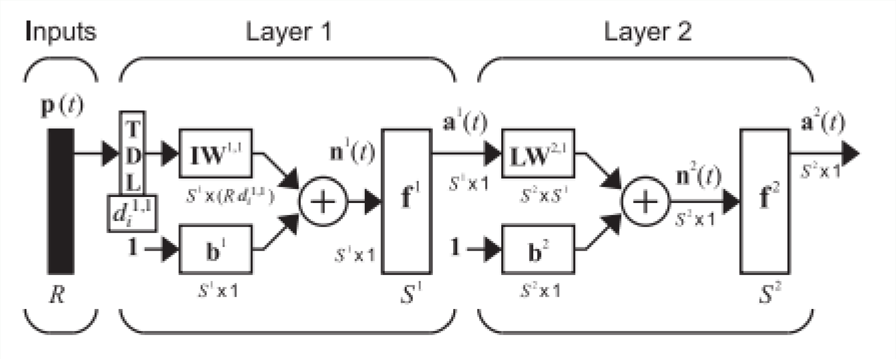
LSTM

Long Short-Term Memory

- Recurrent model with gated memory
- Designed for longer dependencies
- Strong general sequence model
- Included as the more complex benchmark

FTDNN concept

A feedforward network sees history through a tapped delay line at the input.



- The delay line collects recent input values before the first layer.
- Because there is no feedback loop, the gradients can be computed with standard static backpropagation.
- That makes FTDNN a strong “simpler than LSTM” baseline for short-term forecasting.

FINAL MAE

22.449 W

R^2

0.5452

FTDNN result

Nonlinearity made a large difference.

Model	MAE (W)	RMSE (W)	R^2
FTDNN nonlinear	22.449	31.218	0.5452
NARX nonlinear	23.018	32.856	0.4962
LSTM	25.541	33.380	0.4800
NARX linear	28.790	37.510	0.3434
FTDNN linear	30.547	39.566	0.2694

Main takeaway

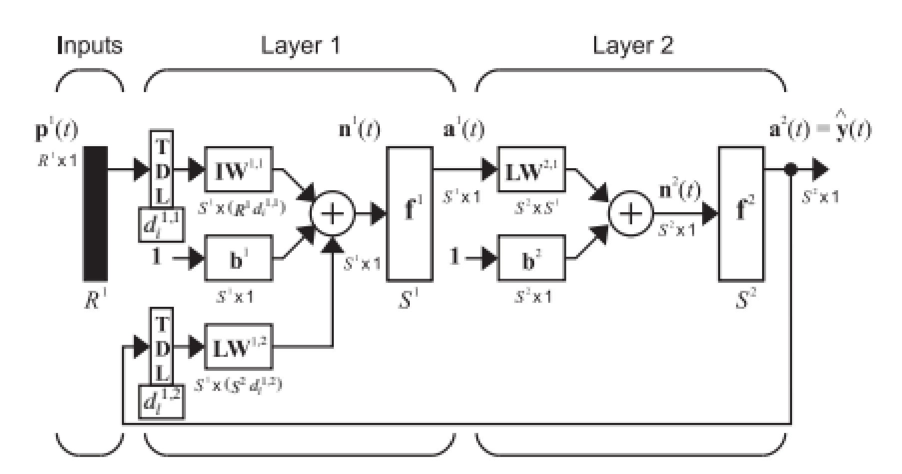
FTDNN nonlinear ranked #1 overall. The linear version was last, so the nonlinearity mattered much more than simply using delayed inputs.

How to explain it

A pure linear delayed-input model can capture trend and lag, but the nonlinear version can model interactions between irradiance, battery state, and recent panel behavior.

NARX concept

NARX combines recent exogenous inputs with recent output history.



- Autoregressive: it looks at previous values of the predicted series.
- Exogenous inputs: it also looks at external signals such as weather and irradiance.
- This makes it a natural baseline for dynamic-system forecasting.

FINAL MAE

23.018 W

R^2

0.4962

NARX result

NARX nonlinear was the runner-up and clearly beat NARX linear.

Model	MAE (W)	RMSE (W)	R^2
FTDNN nonlinear	22.449	31.218	0.5452
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What worked well

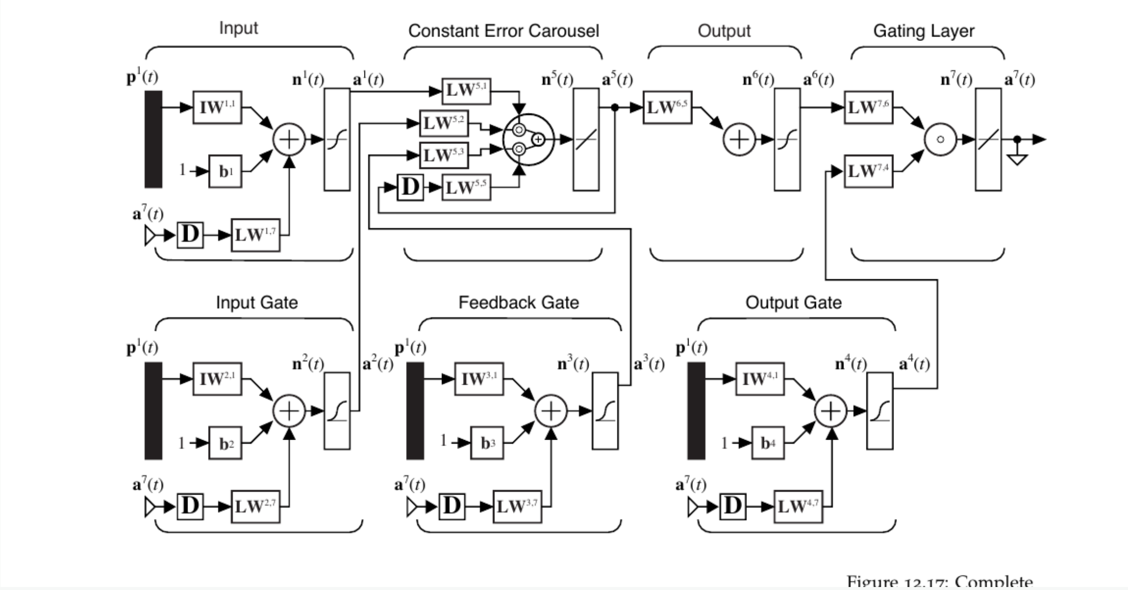
Using past output history helped the model stay competitive with FTDNN, especially on the one-hour horizon.

What limited it

In this run it still trailed the nonlinear FTDNN, suggesting the extra autoregressive structure did not add enough benefit to overcome the simpler model.

LSTM concept

LSTM adds gates around a recurrent memory path to preserve longer dependencies.



- LSTM was designed to maintain useful memory while keeping training stable.
- Its input, forget, and output gates control what information enters, stays in, and leaves the memory state.
- That flexibility often helps when long-term context matters.
- In this project, it worked well but did not beat the best simpler model.

FINAL MAE	R^2
25.541 W	0.4800

LSTM result

The recurrent model was competitive, but not the winner on this dataset.

Model	MAE (W)	RMSE (W)	R^2
FTDNN nonlinear	22.449	31.218	0.5452
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Why LSTM may not have won here

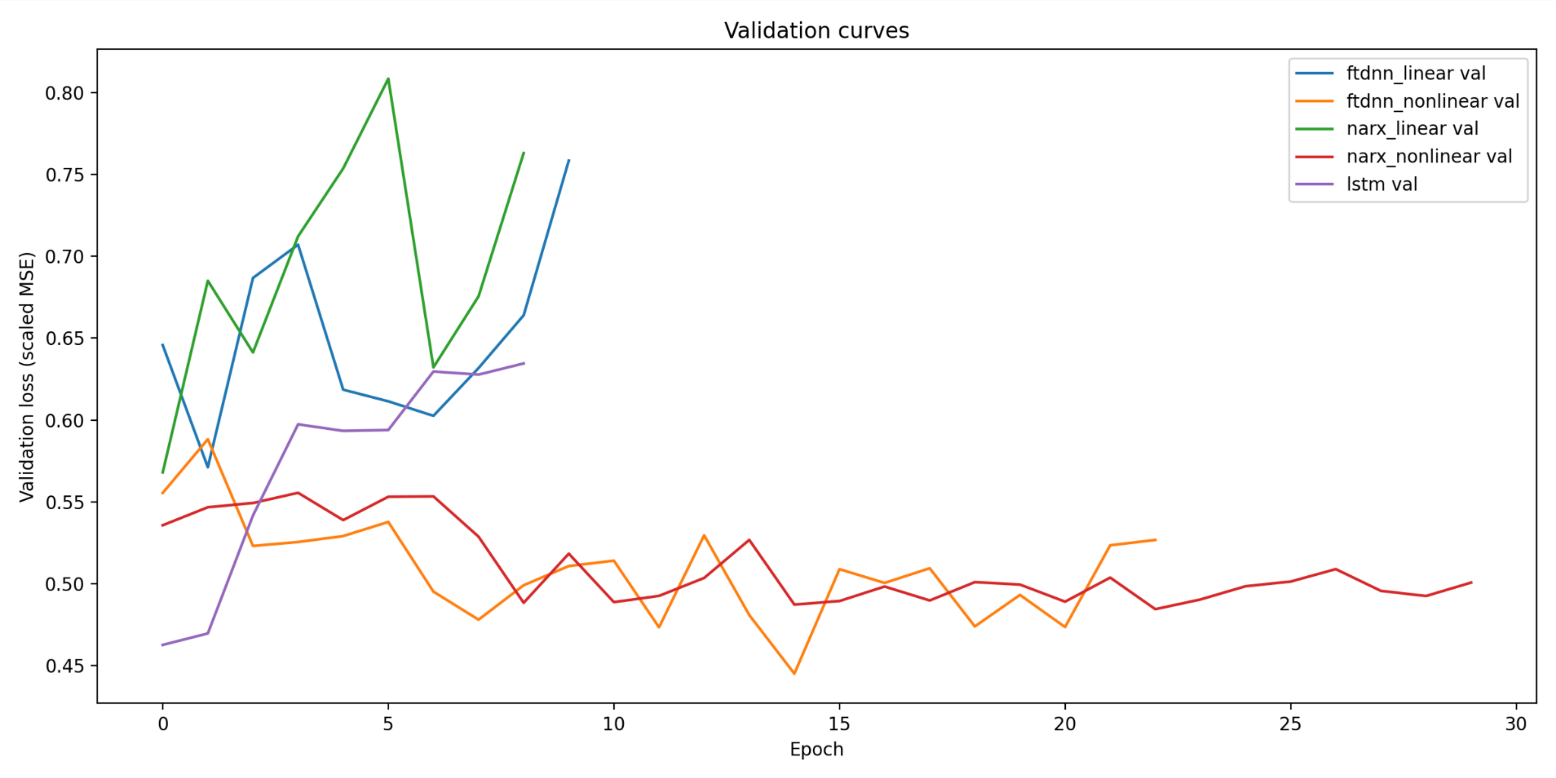
The horizon is only +60 min, so recent local structure may be enough. That favors simpler finite-memory models such as nonlinear FTDNN.

Still a useful finding

A negative result is still valuable: more complex sequence models do not automatically outperform simpler baselines on real data.

Training behavior

Early stopping kept the selected models close to their best validation performance.



Validation takeaway

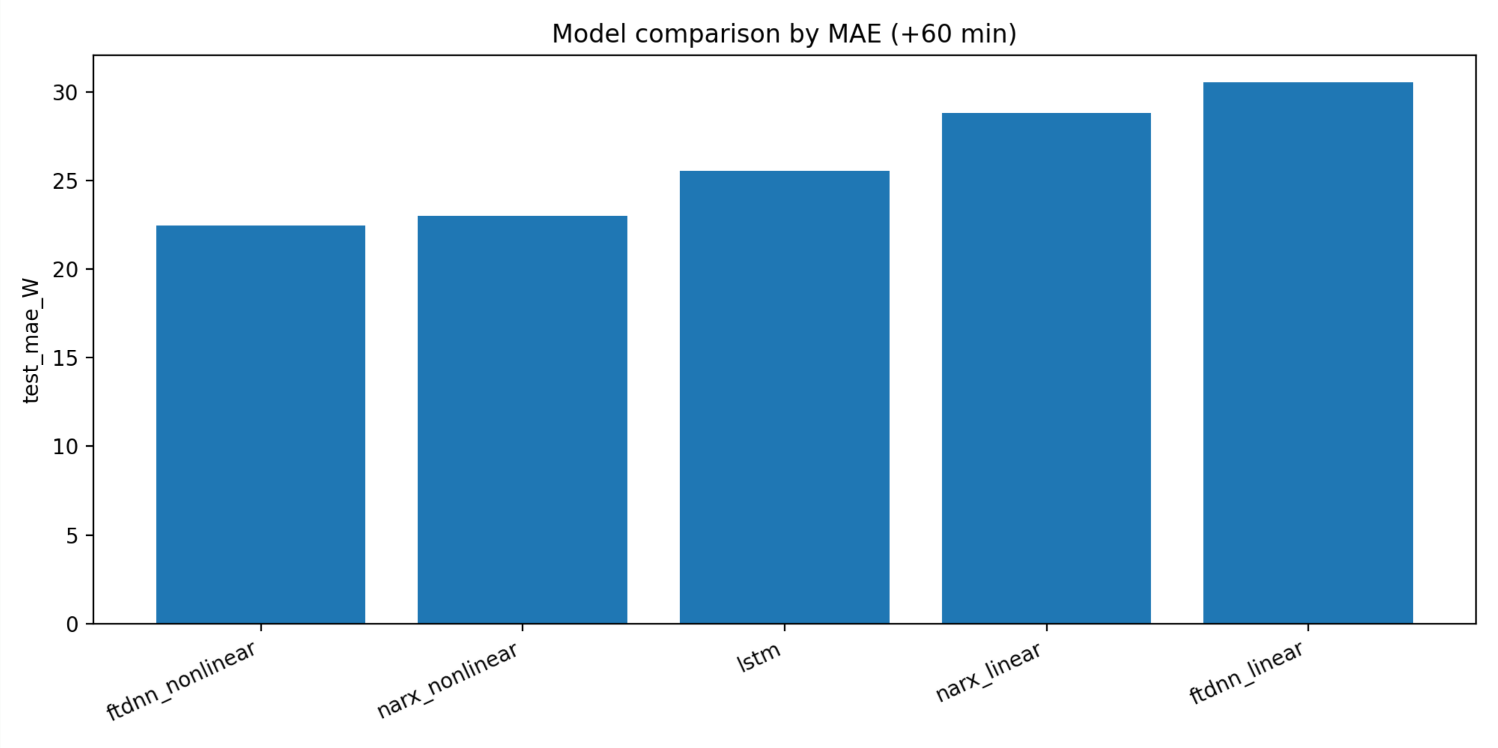
The nonlinear FTDNN and nonlinear NARX curves settled to lower validation losses than the linear variants.

LSTM takeaway

LSTM validated well early, then stopped quickly. It was competitive, but that did not translate into the best test MAE.

Result 1: MAE comparison

Lower MAE means fewer watts of average forecast error.



Best MAE

FTDNN nonlinear
22.449 W

Runner-up

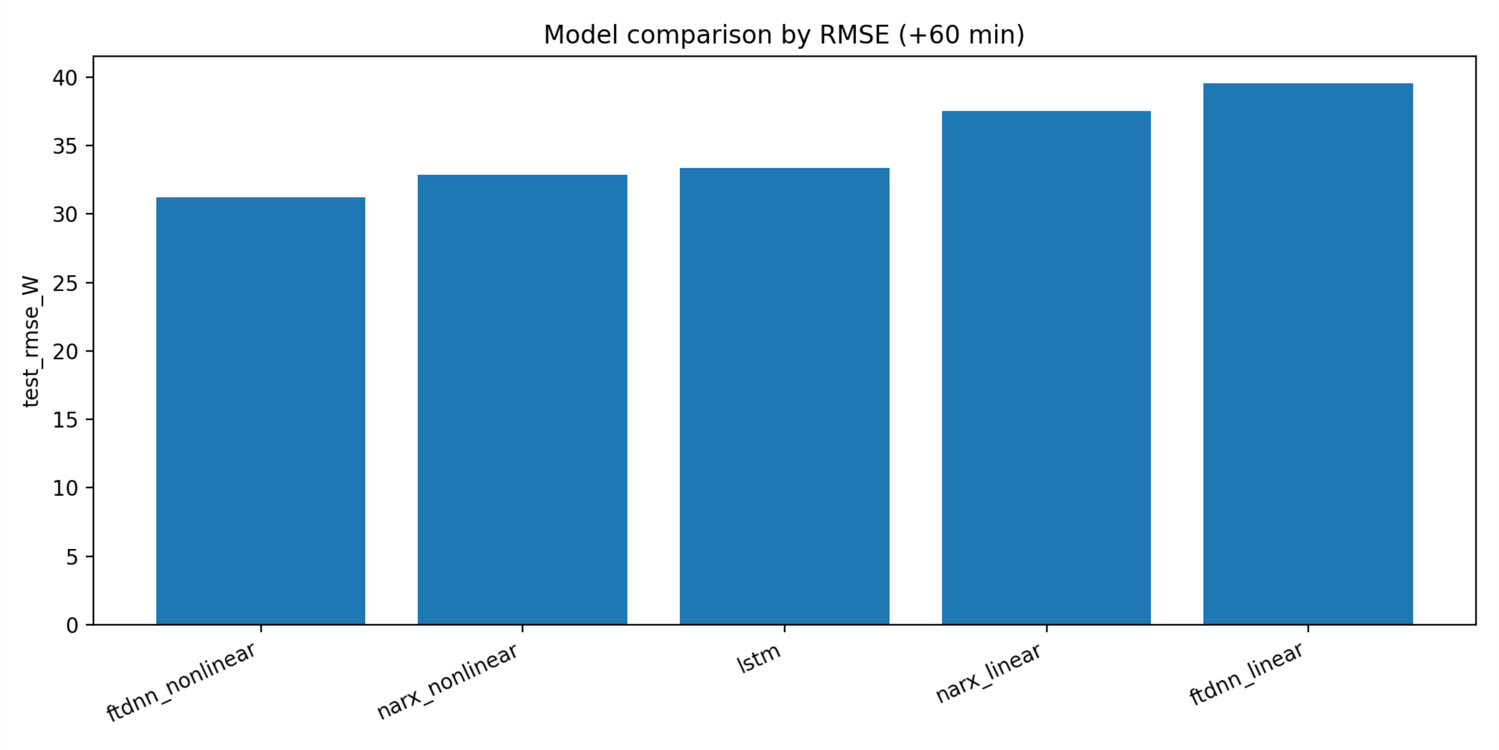
NARX nonlinear
23.018 W

Main message

Both nonlinear baselines beat LSTM.

Result 2: RMSE comparison

RMSE highlights larger misses more strongly than MAE.



Best RMSE

FTDNN nonlinear
31.218 W

Runner-up

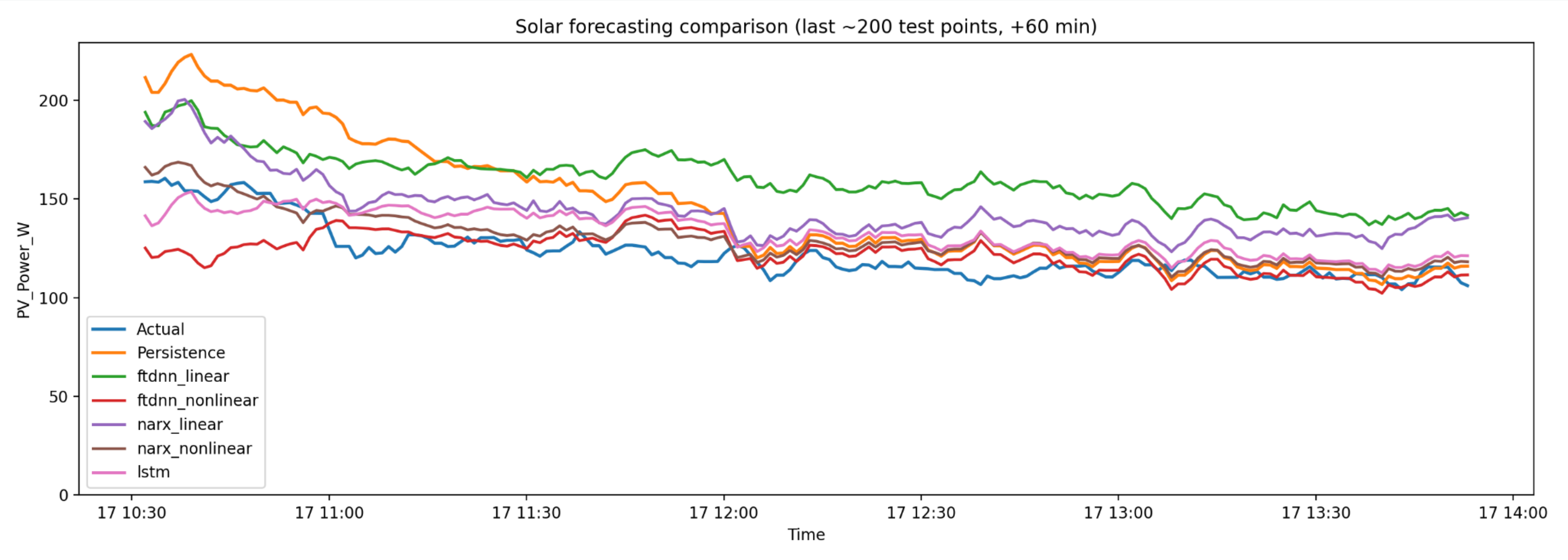
NARX nonlinear
32.856 W

Interpretation

The same ranking holds under a stricter error metric.

Result 3: actual vs predicted on the test set

The final comparison plot makes the ranking visually intuitive.



Conclusions

The best result came from the simpler nonlinear delayed-input model.

1. Best overall model

FTDNN nonlinear achieved the strongest final test performance:

MAE = 22.449 W
RMSE = 31.218 W
R^2 = 0.5452

2. Nonlinearity mattered

For both FTDNN and NARX, the nonlinear version clearly outperformed the linear version. That supports the recommendation to compare against linear baselines first.

3. Bigger was not automatically better

LSTM worked and beat both linear baselines, but it did not beat the best simpler models on this one-hour forecasting problem.

Questions?

Short-Term Solar Power Forecasting • FTDNN vs NARX vs LSTM

Thank you.